

Network Load Balancer and Application Load Balancer Chaining

Combining Layer 4 Performance with Layer 7 Intelligent Routing



Why NLB Alone Is Not Enough

LAYER 4 — TRANSPORT LAYER

Strengths

→ High Performance

Handles millions of requests per second with ultra-low latency.

→ Static IP Support

Assigns fixed IPs per AZ — ideal for firewall whitelisting.

→ Low Latency

Operates at the connection level without inspecting payloads.

Limitations

→ No Path-Based Routing

Cannot differentiate `/api` from `/app` traffic.

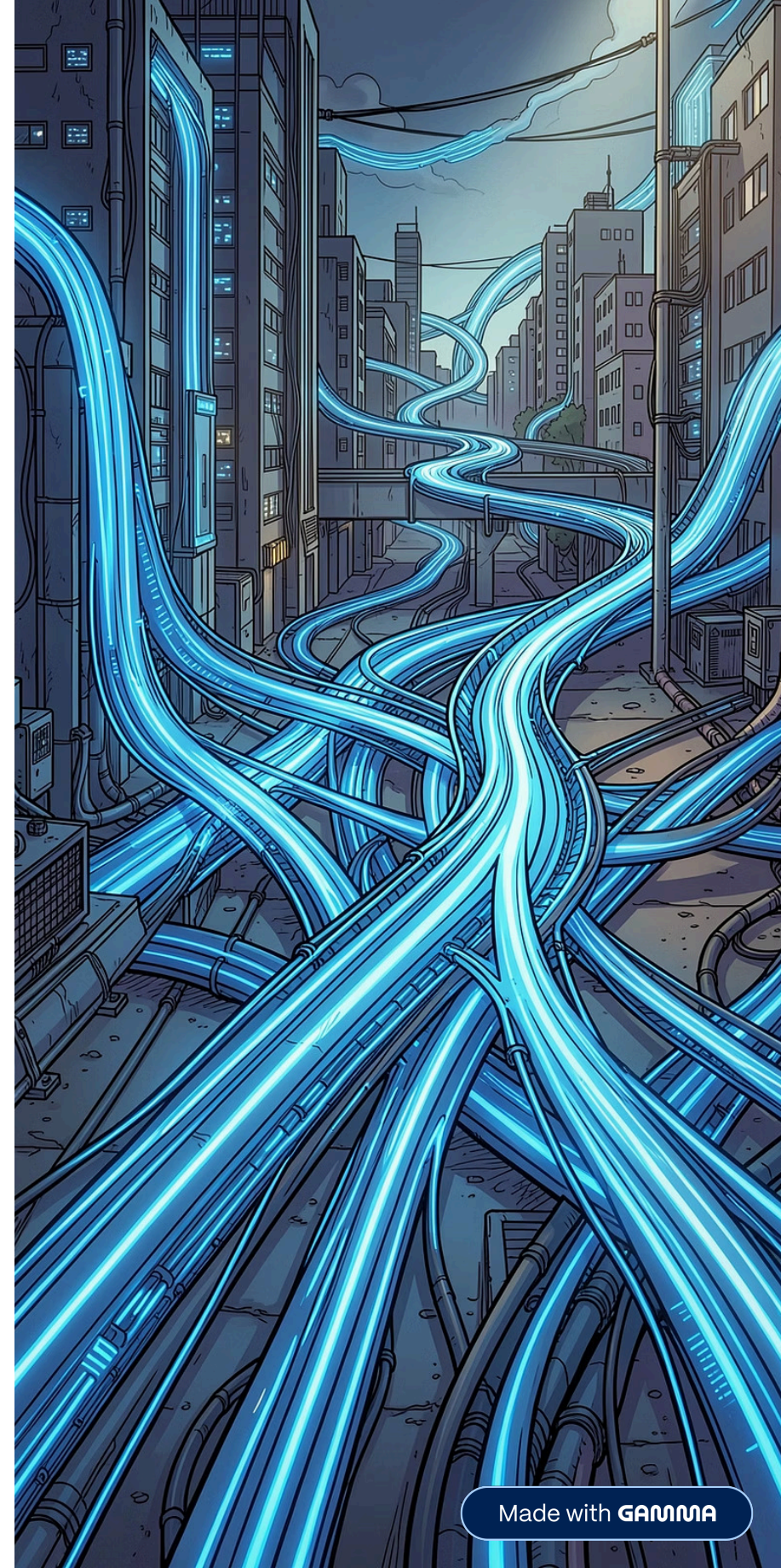
→ No URL Inspection

Operates only on IP/port — unaware of HTTP headers or paths.

→ No HTTP Awareness

Cannot make routing decisions based on application-layer data.

i NLB handles network traffic efficiently, but modern applications require intelligent application-level routing.



Why Not Use Only ALB?

LAYER 7 — APPLICATION LAYER

Strengths

→ Path-Based Routing

Routes `/api/*` and `/app/*` to separate target groups.

→ HTTP/HTTPS Aware

Inspects headers, host names, query strings, and methods.

→ Smart Distribution

Supports weighted routing, sticky sessions, and redirects.

Limitations

→ No Static IP

ALB DNS resolves to dynamic IPs — problematic for whitelisting.

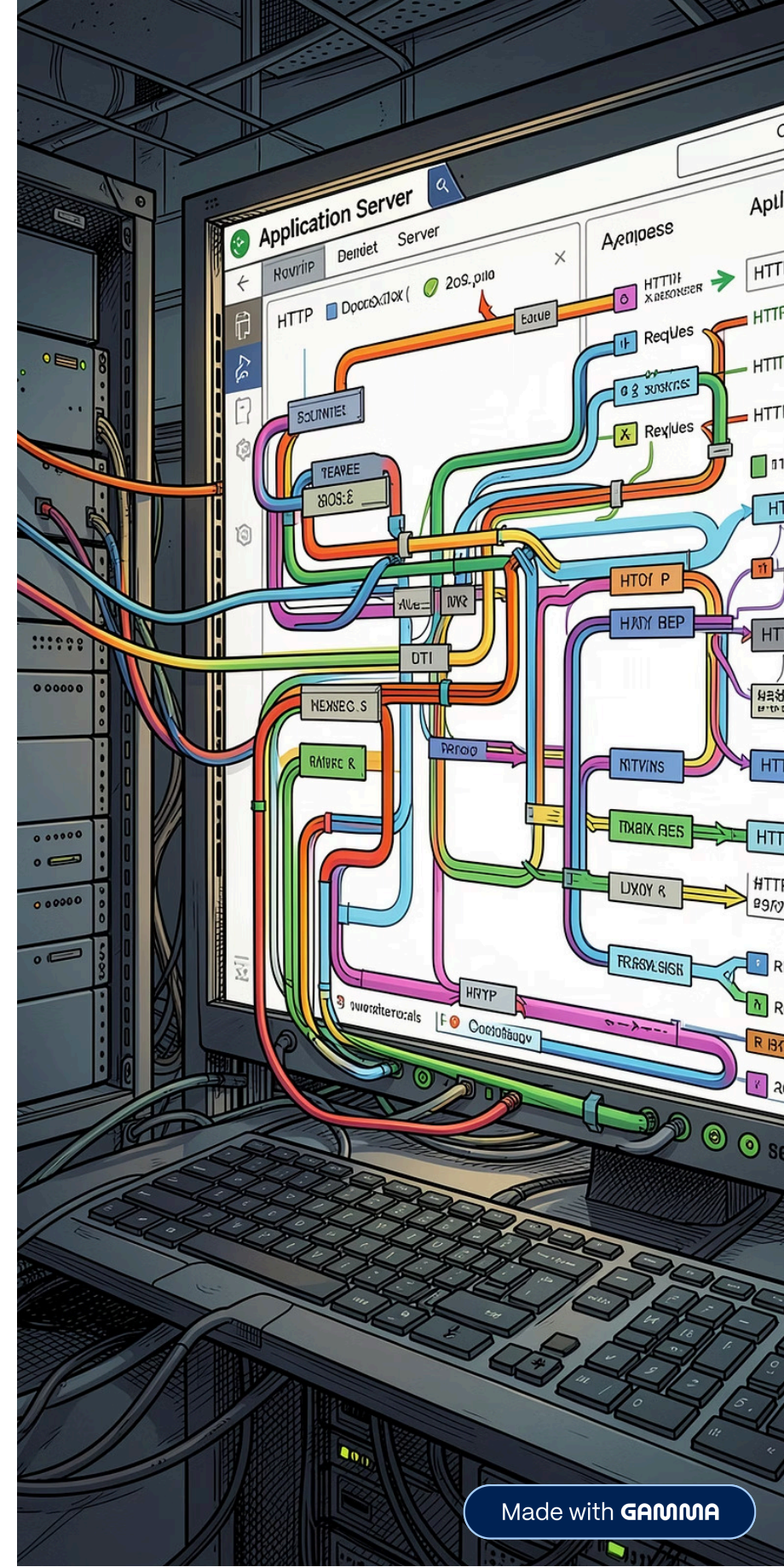
→ No Raw TCP Support

Cannot handle non-HTTP protocols or raw TCP connections.

→ Firewall Challenges

Dynamic IPs make external firewall rule management difficult.

i ALB is intelligent for application traffic, but lacks critical Layer 4 networking advantages.



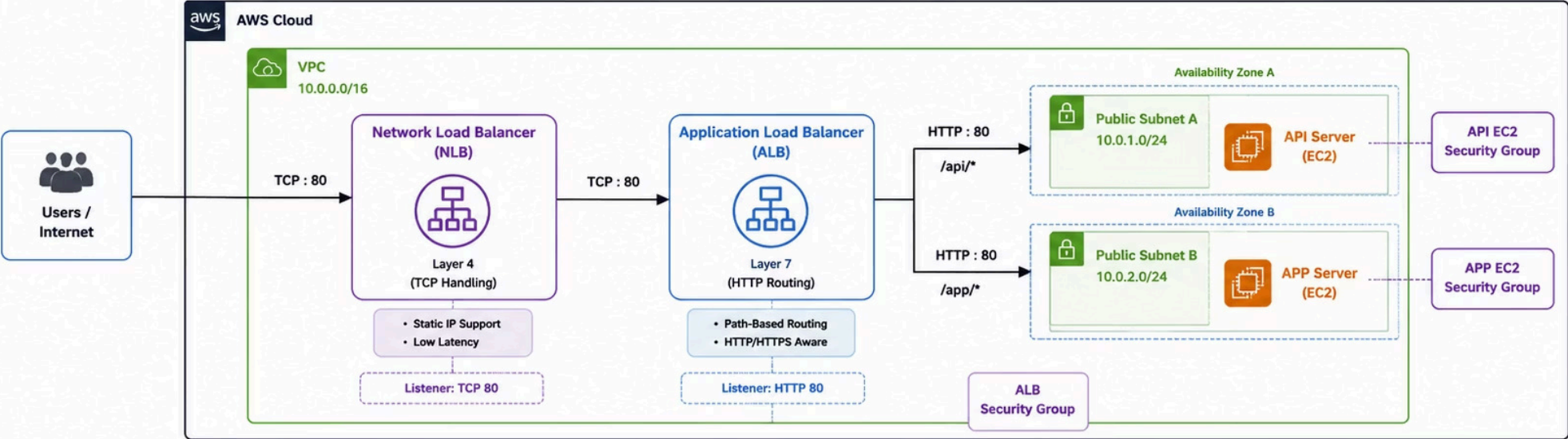
Why Combine NLB and ALB?

Enterprise Requirement	Solution
Static IP	NLB
Low latency	NLB
TCP handling	NLB
Path-based routing	ALB
HTTP awareness	ALB
Intelligent routing	ALB

⚠ Modern enterprise applications require both Layer 4 performance and Layer 7 intelligent routing.

NLB → ALB → EC2 Architecture

High Availability Across Availability Zones



Security Groups & Ports

1. NLB (Network Load Balancer)
NLB does not use Security Groups

2. ALB Security Group			
Type	Port	Source	Description
HTTP	80	0.0.0.0/0	Allow HTTP from Internet
Outbound Rules			
Type	Port	Destination	Description
HTTP	80	To EC2 SGs	Allow traffic to EC2 instances

3. API EC2 Security Group			
Inbound Rules			
Type	Port	Source	Description
HTTP	80	ALB Security Group	Allow HTTP from ALB
SSH	22	My IP	Allow SSH for access
Outbound Rules			
Type	Port	Destination	Description
All Traffic	All	0.0.0.0/0	Allow all outbound

4. APP EC2 Security Group			
Inbound Rules			
Type	Port	Source	Description
HTTP	80	ALB Security Group	Allow HTTP from ALB
SSH	22	My IP	Allow SSH for access
Outbound Rules			
Type	Port	Destination	Description
All Traffic	All	0.0.0.0/0	Allow all outbound

Legend (Traffic Flow)

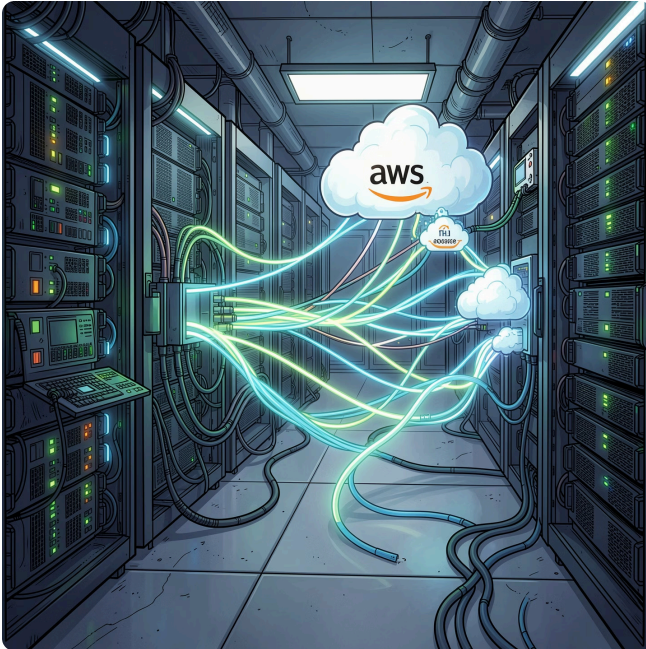
- TCP : 80 (Internet to NLB)
- TCP : 80 (NLB to ALB)
- HTTP : 80 (ALB to EC2)

Key Points

- NLB provides static IP and high performance (Layer 4)
- ALB provides intelligent routing based on URL paths (Layer 7)
- Architecture is highly available across multiple AZs

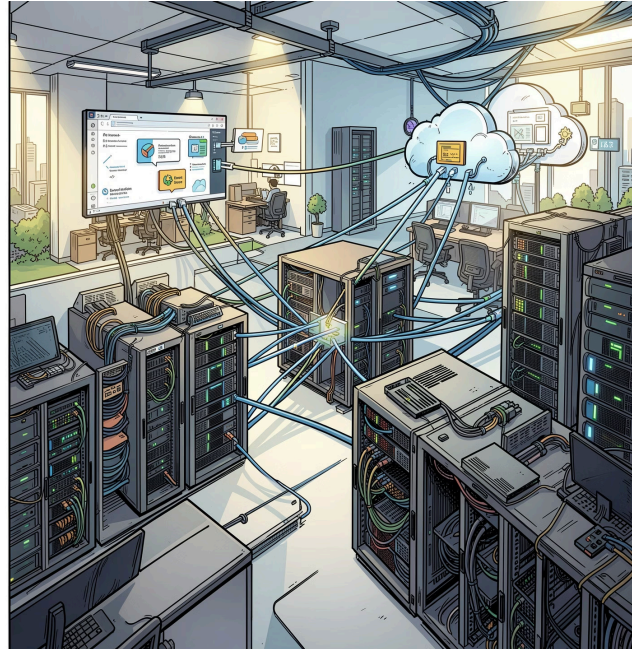
Real-World Use Cases

How private connectivity, hybrid networking, and API routing work together in practical enterprise environments.



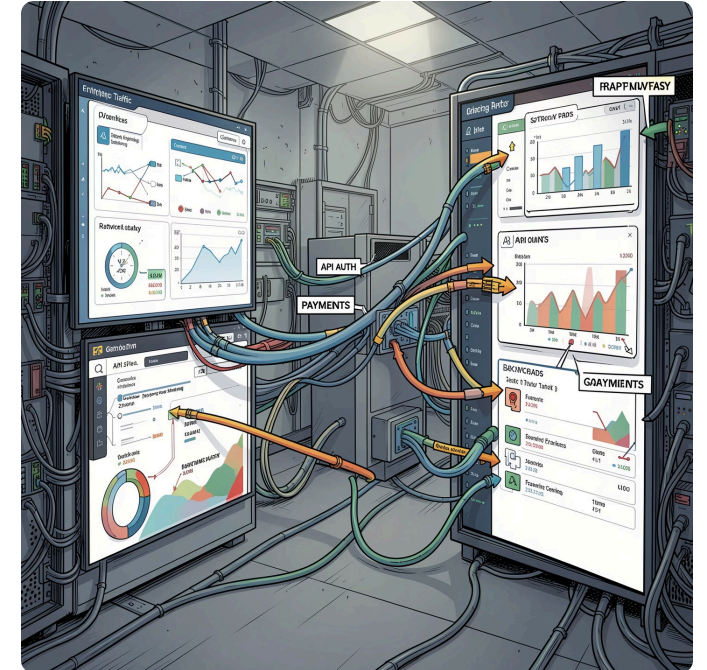
AWS PrivateLink

- Secure private connectivity without internet exposure
- Uses NLB as the service endpoint
- ALB can perform internal application routing



Hybrid Enterprise Networks

- On-premise systems connect securely to cloud applications
- NLB provides stable Layer 4 connectivity
- ALB handles intelligent HTTP routing internally



Enterprise API Platforms

- External partner systems connect using fixed endpoints
- NLB acts as the public network entry point
- ALB routes requests to backend services like: /api, /auth, /payments

✔ Key takeaway: NLB and ALB complement each other by combining stable private connectivity with intelligent application routing.